

## Exercise 2: E-commerce Platform Search Function

```
PS C:\Users\HP\GitDemo\CTS_HANDSON\Week1_Deepskilling\Algorithms_DataStructures\EcommerceSearch> & 'C:\Program Files\Java\jdk-11\bin\java.exe' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\72a6e29b1c1b32c7695024d82dc2799\redhut.java\jdt_ws\EcommerceSearch_aea42f79\bin' 'SearchProduct'
```

Linear Search Result:  
ID: 102  
Name: Phone  
Category: Electronics

Binary Search Result:  
ID: 102  
Name: Phone

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ID: 102  
Name: Phone  
Category: Electronics

Time Complexity Analysis:  
Linear Search - Best:  $O(1)$ , Average:  $O(n)$ , Worst:  $O(n)$   
Binary Search - Best:  $O(1)$ , Average:  $O(\log n)$ , Worst:  $O(\log n)$   
Binary Search is more suitable for large sorted datasets because it reduces the search space by half in every step.

PS C:\Users\HP\GitDemo\CTS\_HANDSON\Week1\_Deepskilling\Algorithms\_DataStructures\EcommerceSearch>

## Exercise 7: Financial Forecasting

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR
```

Run: FinancialForecast

```
PS C:\Users\HP\GitDemo\CTS_HANDSON\Week1_Deepskilling\Algorithms_DataStructures\FinancialForecasting> & 'C:\Program Files\Java\jdk-11\bin\java.exe' '-cp' 'C:\Users\HP\AppData\Roaming\Code\User\workspaceStorage\2da789bd4599ddeff3a2e61f92b41a15\redhut.java\jdt_ws\FinancialForecasting_250e3aed\bin' 'FinancialForecast'
```

Future Value after 5 years = 16185.10

PS C:\Users\HP\GitDemo\CTS\_HANDSON\Week1\_Deepskilling\Algorithms\_DataStructures\FinancialForecasting>